

Constraint-Aware Flow Matching via Randomized Exploration

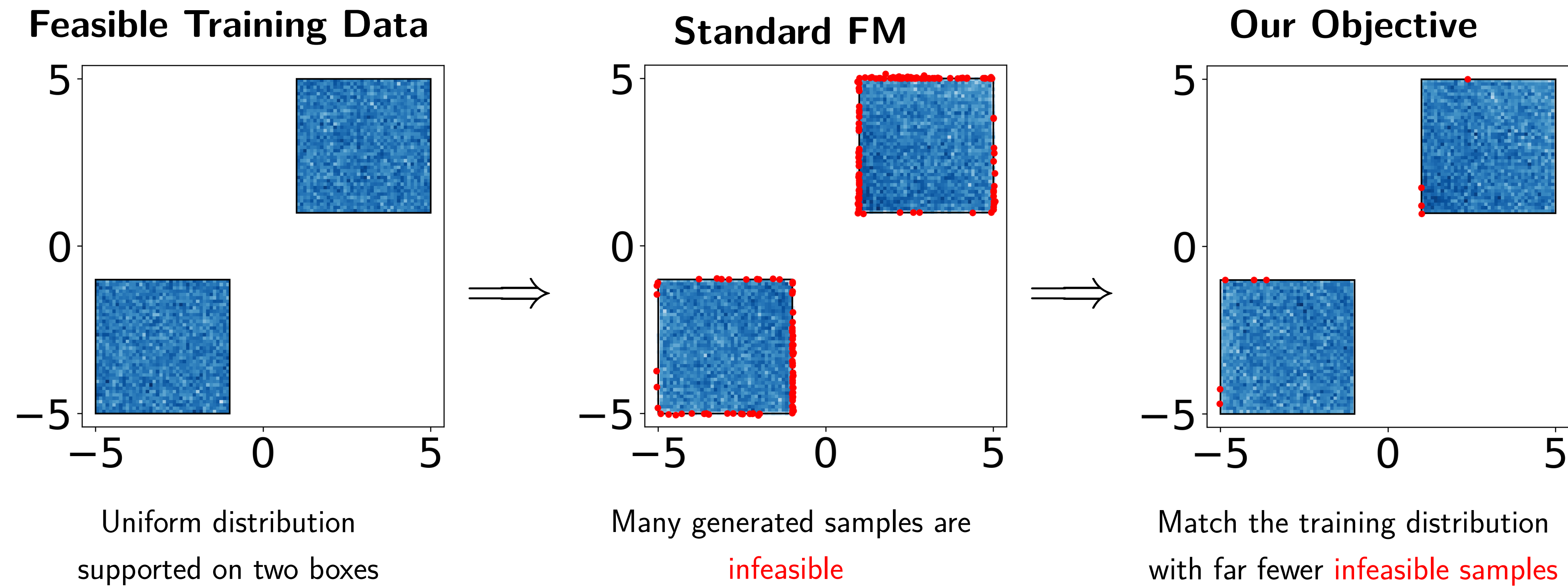
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Constraint-Aware Generation

- **Motivation and Problem:** even when trained exclusively on feasible samples in the constraint set $\mathcal{C}$, generative models may generate samples that violate the constraints, i.e., $x_{\text{gen}} \notin \mathcal{C}$.



- **Existing Work:** Mirror-Diffusion Models require $\mathcal{C}$ to be convex; Neural Approximate Mirror Maps require an available distance function to $\mathcal{C}$; and Projected Diffusion Models require access to a projection onto $\mathcal{C}$.

- **Our Work:** ONLY access to a membership oracle for $\mathcal{C}$ is required:

$$\mathbf{1}_{\mathcal{C}}(x) = \begin{cases} 1, & x \in \mathcal{C}, \\ 0, & x \notin \mathcal{C}. \end{cases}$$

- **Why oracle $\mathcal{C}$ matters?** Stable molecule generation; Physical trajectories generation following laws, etc.
- **Objective:** Let q_1 denote the training-data distribution, with $\text{supp}(q_1) \subseteq \mathcal{C}$. Learn a flow matching (FM) velocity field $u_{\theta} : \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$ such that

$$X_0^{\theta} = X_0 \sim \mathcal{N}(0, I), \quad \frac{dX_t^{\theta}}{dt} = u_{\theta}(X_t^{\theta}, t),$$

and the generated samples X_1^{θ} approximately follow q_1 while satisfying $\mathbb{P}(X_1^{\theta} \in \mathcal{C}) \approx 1$.

- **Methodology:**

$$\arg \min_{\theta} \left\{ \underbrace{\int_0^1 \mathbb{E} [\|u_{\theta}(X_t, t) - (X_1 - X_0)\|^2] dt}_{\text{standard FM objective}} - \underbrace{\lambda}_{\text{constraint weight}} \cdot \underbrace{\mathbb{E}[\mathbf{1}_{\mathcal{C}}(X_1^{\theta})]}_{\text{constraint-aware objective}} \right\}, \quad (1)$$

in which $X_t = (1 - t)X_0 + tX_1$ and $X_1^{\theta} = X_0 + \int_0^1 u_{\theta}(X_t^{\theta}, t) dt$.

- **Challenge:** exact membership oracle provides no useful gradient. The indicator is constant away from the boundary of $\mathcal{C}$:

$$\nabla_x \mathbf{1}_{\mathcal{C}}(x) = \begin{cases} 0, & x \notin \partial \mathcal{C}, \\ \text{undefined}, & x \in \partial \mathcal{C}. \end{cases}$$

Flow Matching via Random Exploration (FM-RE)'s Key Idea: Randomized Velocity

Gaussian exploration is introduced into the velocity:

$$\tilde{U}_t^{\theta, \sigma} = u_{\theta}(X_t^{\theta, \sigma}, t) + \underbrace{\sigma_t W}_{\text{randomized exploration}}, \quad W \sim \mathcal{N}(0, I), \quad (2)$$

where σ_t controls the exploration scale. With Euler discretization (step size Δt), $X_t^{\theta, \sigma}$ follows

$$X_{t+\Delta t}^{\theta, \sigma} = X_t^{\theta, \sigma} + \tilde{U}_t^{\theta, \sigma} \Delta t, \quad X_0^{\theta, \sigma} = X_0.$$

This leads to the access of PDF of $\tilde{U}_t^{\theta, \sigma} \in \mathbb{R}^d$ based on $x_t^{\theta, \sigma}$: $\pi_{\theta, \sigma}(\tilde{U}_t^{\theta, \sigma} | x_t^{\theta, \sigma}) = \mathcal{N}(\tilde{U}_t^{\theta, \sigma} | u_{\theta}(x_t^{\theta, \sigma}, t), \sigma_t^2 I)$.

Tractable Gradients under Randomized Velocity

Under randomized velocity in (2), let $\mathcal{T}$ denote an N -step generation trajectory and $X_1(\mathcal{T})$ denote the trajectory's induced sample, i.e.,

$$\mathcal{T} = (X_0^{\theta, \sigma}, \tilde{U}_0^{\theta, \sigma}, X_{\Delta t}^{\theta, \sigma}, \tilde{U}_{\Delta t}^{\theta, \sigma}, X_{2\Delta t}^{\theta, \sigma}, \tilde{U}_{2\Delta t}^{\theta, \sigma}, \dots, X_1^{\theta, \sigma}), \quad X_1(\mathcal{T}) = X_1^{\theta, \sigma}.$$

The gradient of $\mathbb{E}[\mathbf{1}_{\mathcal{C}}(X_1^{\theta, \sigma})]$ becomes tractable,

$$\nabla_{\theta, \sigma} \mathbb{E}[\mathbf{1}_{\mathcal{C}}(X_1^{\theta, \sigma})] = \mathbb{E}_{\mathcal{T}} \left[\sum_{i=0}^{N-1} \left(\nabla_{\theta, \sigma} \log \pi_{\theta, \sigma}(\tilde{U}_{i\Delta t}^{\theta, \sigma} | X_{i\Delta t}^{\theta, \sigma}) \right) \mathbf{1}_{\mathcal{C}}(X_1(\mathcal{T})) \right].$$

Efficient two-stage training. Choose a randomization start time $t_0 = N_1 \Delta t$, so that only the final $N_2 = N - N_1$ steps are randomized.

- **Stage 1:** Train a standard FM velocity u_{θ_1} and freeze θ_1 . It deterministically generates for $t \in [0, t_0]$.
- **Stage 2:** Initialize $\theta_2 \leftarrow \theta_1$ and jointly optimize θ_2 and σ for $t \in [t_0, 1]$ with the objective in (1). The resulting velocity **during training** is

$$\tilde{U}_t = \begin{cases} u_{\theta_1}(X_t, t), & 0 \leq t < t_0, \\ u_{\theta_2}(X_t, t) + \sigma_t W, & t_0 \leq t \leq 1, \quad W \sim \mathcal{N}(0, I). \end{cases}$$

- **During sampling:** u_{θ_1} is used for $t < t_0$. And the **mean** velocity u_{θ_2} (without randomization) is used for $t \geq t_0$.

Experimental Results

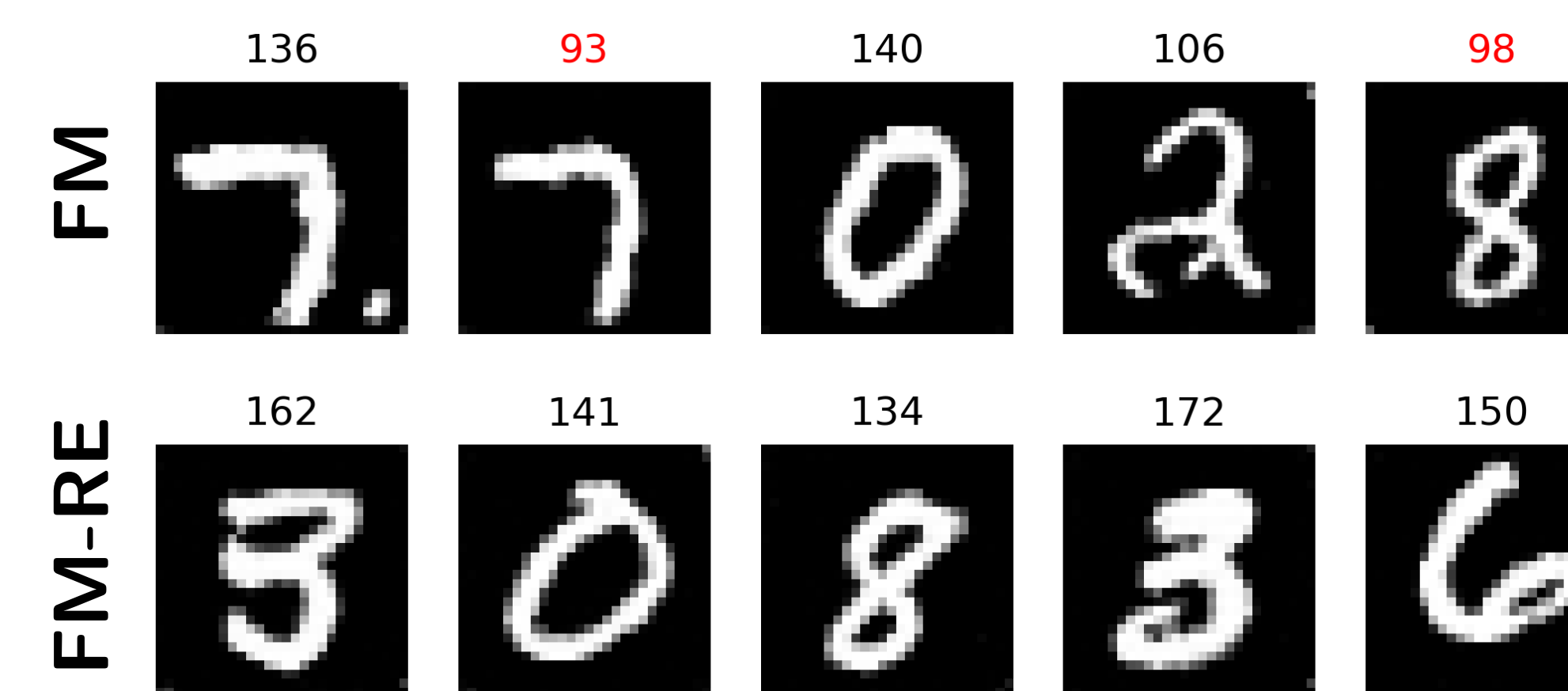
MNIST generation with attribute constraints.

- Train on only feasible samples.
- Aim to generate MNIST images satisfying the prescribed constraints $\mathcal{C}$.

Brightness Constraint

$$\#\{\text{white pixels in } x\} \geq 100$$

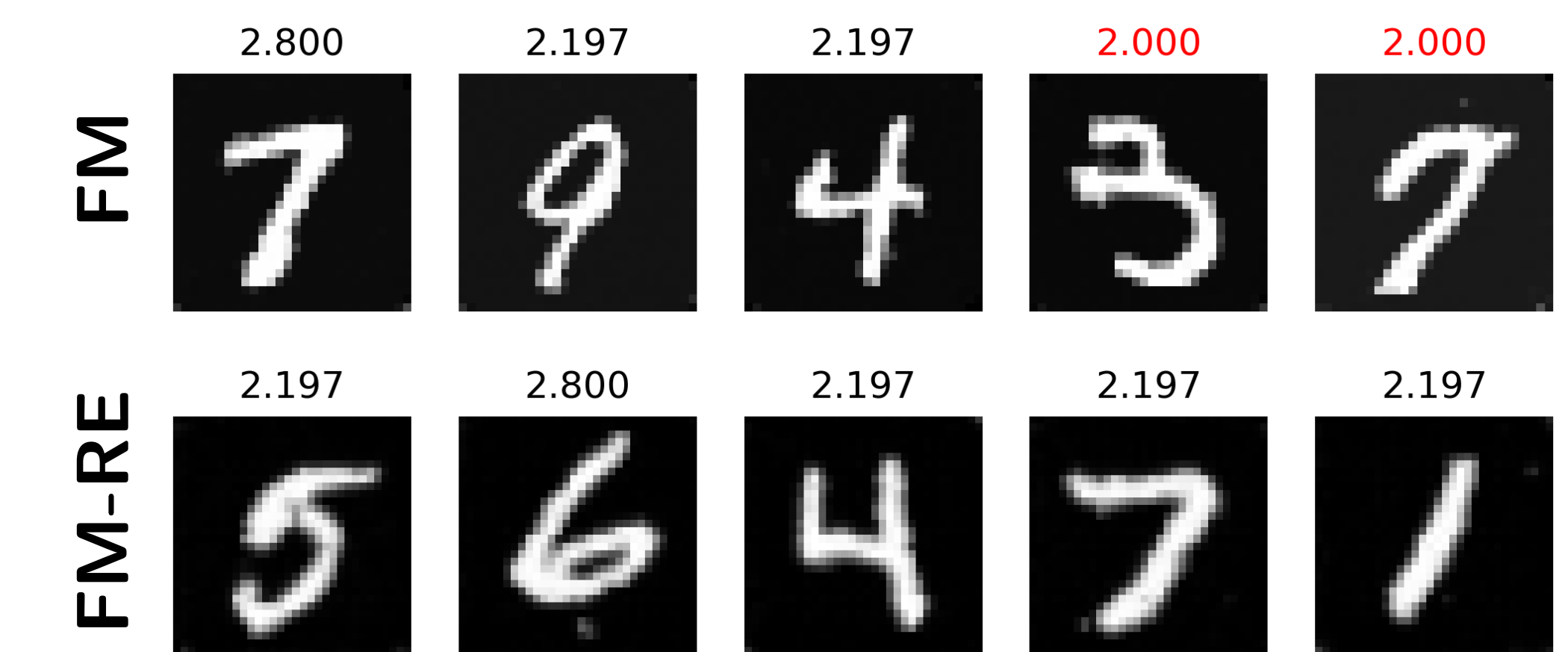
Objective: Generate sufficiently bright digits containing at least 100 white pixels (pixel value > 128).



Thickness Constraint

$$2 < \text{Thickness}(x) < 3$$

Objective: Generate digits whose maximum stroke thickness lies within the prescribed range.



Method	Brightness	Thickness		
	Violation (%)	FID	Violation (%)	FID
Standard FM	9.14	6.16	23.2	6.19
FM-RE (ours)	1.12	5.86	5.40	10.8

Takeaway:

1. FM-RE handles broadly applicable **membership-oracle $\mathcal{C}$** with minimal assumptions on the structure of $\mathcal{C}$.
2. FM-RE achieves a **good trade-off between constraint satisfaction and sample quality**, substantially reducing constraint violations while generating high-quality samples.